The Practise Ground

Class 11-12 Maths

Differentiation Rules

Four rules and the standard derivatives

1

$$\frac{d}{dx}(x^n) = n\,x^{(n-1)}$$

Power rule
Bring the power down, reduce it by 1

2

$$\frac{d}{dx}(uv) = u\,v' + v\,u'$$

Product rule
Derivative of a product of two functions

3

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\,u' - u\,v'}{v^2}$$

Quotient rule
Derivative of a ratio of two functions

4

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Chain rule
Differentiate the outer function, then the inner

Standard results

$\frac{d}{dx}(\sin x) = \cos x$

$\frac{d}{dx}(\cos x) = -\sin x$

$\frac{d}{dx}(e^x) = e^x$ $\frac{d}{dx}(\ln x) = \frac{1}{x}$

Quick examples

$y = x^3 + 2x$ gives $\frac{dy}{dx} = 3x^2 + 2$

$y = (3x + 1)^2$ gives $\frac{dy}{dx} = 2(3x + 1) \times 3 = 6(3x + 1)$

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